Lecture 3: Choice Under Uncertainty

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1. From Deterministic Outcomes to Lotteries

So far the consumer's choice object has been a deterministic bundle $x = (x_1, \ldots, x_n)$. But many real decisions involve uncertainty: buying stocks, purchasing insurance, choosing a used car of unknown quality. We need to enlarge the choice object.

A finite set $A = \{a_1, a_2, \ldots, a_n\}$ of mutually exclusive **outcomes**. Each a_i may itself be a consumption bundle, a monetary payoff, or any object of preference.

Examples: weather states $\{\text{sunny}, \text{rainy}\}$; monetary returns $\{0, 100, 200, 300\}$; outcomes of a coin flip $\{+1, -1\}$.

A **simple lottery** is a probability distribution $P = (p_1, \ldots, p_n)$ over A with

$$p_i \in [0,1] \quad \forall i, \qquad \sum_{i=1}^{n} p_i = 1, \qquad P(a_i) = p_i.$$

We write the lottery as $g = (p_1 \circ a_1,\ p_2 \circ a_2,\ \ldots,\ p_n \circ a_n)$.

Example (fair coin paying \$1 or losing \$1): $g = \big(\tfrac{1}{2} \circ 1,\ \tfrac{1}{2} \circ (-1)\big)$.

A **compound lottery** is a lottery whose outcomes are themselves lotteries:

$$g \;=\; \big(\alpha \circ P,\, (1-\alpha) \circ P'\big), \qquad P,\, P' \in \mathcal{L}.$$

where $\mathcal{L}$ denotes the space of all lotteries (simple and compound).

There is no limit to how deeply a lottery can be composed — but at every leaf of the composition tree there must be an outcome in A.

2. Preferences Over Lotteries

The consumer has a preference relation $\succsim$ over $\mathcal{L}$. We carry over the two basic axioms from Lecture 1 and add four more tailored to risky alternatives.

2.1. Standard Axioms

For all $g, g' \in \mathcal{L}$: $g \succsim g'$ or $g' \succsim g$.

For all $g, g', g'' \in \mathcal{L}$: if $g \succsim g'$ and $g' \succsim g''$, then $g \succsim g''$.

Since $\mathcal{L}$ contains the degenerate lotteries $(1 \circ a_i)$, A1–A2 imply we can rank the underlying outcomes

$a_1 \succsim a_2 \succsim \cdots \succsim a_n$.

2.2. Axioms Specific to Risk

For every $g \in \mathcal{L}$ there exists $\alpha \in [0,1]$ such that

$$g \sim \big(\alpha \circ a_1,\, (1-\alpha) \circ a_n\big),$$

where a_1 and a_n are the best and worst outcomes in A.

Example. If $A = \{\$1000,\, \$10,\, \text{death}\}$ ranked from best to worst, A3 says there exists some probability α for which the agent is indifferent between \$10 for sure and a lottery paying \$1000 with probability α and death with probability $1-\alpha$.

For all $\alpha, \beta \in [0,1]$:

$$\big(\alpha \circ a_1,\, (1-\alpha) \circ a_n\big) \succsim \big(\beta \circ a_1,\, (1-\beta) \circ a_n\big) \iff \alpha \geq \beta.$$

This requires $a_1 \succ a_n$ — it eliminates indifference between best and worst outcomes.

If $g = (p_1 \circ g^1, \ldots, p_k \circ g^k)$ and $h = (p_1 \circ h^1, \ldots, p_k \circ h^k)$ with $h^i \sim g^i$ for every i, then $h \sim g$.

Interpretation: If you're indifferent between two lotteries that show up with the same probability inside a bigger lottery, you should be indifferent between the bigger lotteries too.

For any $g \in \mathcal{L}$, if $(p_1 \circ a_1, \ldots, p_n \circ a_n)$ is the simple lottery on A induced by g (computed by collapsing all the compound stages), then

$$(p_1 \circ a_1, \ldots, p_n \circ a_n) \sim g.$$

A6 says the agent only cares about the *final* probabilities assigned to each outcome — not about how the lottery is presented or staged. It removes any preference for compound structure per se.

Reducing a compound lottery. Suppose $A = \{a_1, a_2\}$ and

$$g = \big(\alpha \circ a_1,\, (1-\alpha) \circ g'\big), \qquad g' = \big(\beta \circ a_1,\, (1-\beta) \circ a_2\big).$$

Then the reduced simple lottery is

$$P(a_1) = \alpha + (1-\alpha)\beta, \qquad P(a_2) = (1-\alpha)(1-\beta).$$

3. The Von Neumann–Morgenstern Expected Utility Theorem

A utility function $u : \mathcal{L} \to \mathbb{R}$ has the **expected utility property** if, for every lottery g with induced simple lottery $(p_1 \circ a_1, \ldots, p_n \circ a_n)$,

$$u(g) \;=\; \sum_{i=1}^{n} p_i \, u(a_i).$$

The utility of a lottery equals the *expected value of the utility of its outcomes*.

Let $\succsim$ be a preference relation on $\mathcal{L}$ satisfying axioms A1–A6. Then there exists a utility function $u : \mathcal{L} \to \mathbb{R}$ that

1. represents $\succsim$: $u(g) \geq u(g') \iff g \succsim g'$;
2. has the expected utility property.

Proof sketch. For each lottery $g \in \mathcal{L}$ define $u(g) \in [0,1]$ via the indifference

$$g \;\sim\; \big(u(g) \circ a_1,\, (1-u(g)) \circ a_n\big).$$

A3 (continuity) guarantees existence of such a number and A4 (monotonicity) guarantees uniqueness.

Representation. Take $g, g' \in \mathcal{L}$. By construction $g \sim (u(g) \circ a_1,\, (1-u(g)) \circ a_n)$ and similarly for g'. Transitivity and A4 give $g \succsim g' \iff u(g) \geq u(g')$.

Expected utility property. Let $g_S = (p_1 \circ a_1, \ldots, p_n \circ a_n)$ be the simple lottery induced by g. By A6, $g \sim g_S$, so it suffices to verify the property on g_S. For each outcome,

$$a_i \;\sim\; \big(u(a_i) \circ a_1,\, (1-u(a_i)) \circ a_n\big).$$

By A5 (substitution), replacing each a_i inside g_S by its equivalent two-outcome lottery and then reducing (A6) gives

$$g_S \;\sim\; \Big(\Big(\textstyle\sum_i p_i \, u(a_i)\Big) \circ a_1,\, \Big(1 - \textstyle\sum_i p_i \, u(a_i)\Big) \circ a_n\Big).$$

By the definition of u, the coefficient of a_1 on the right-hand side equals $u(g_S)$, hence $u(g_S) = \sum_i p_i \, u(a_i)$. ■

3.1. Numerical Construction of vNM Utility

Example

Let $A = \{\$0,\ \$5,\ \$10\}$ with ranking $\$10 \succ \$5 \succ \$0$.

Step 1. Normalize: assign $u(\$10) = 1$ and $u(\$0) = 0$.
Step 2. Use A3 to find $p_0 \in [0, 1]$ such that

$$\$5 \ \sim\ \big(p_0 \circ \$10,\ (1 - p_0) \circ \$0\big).$$

By the expected utility property,

$$u(\$5) \ =\ p_0\, u(\$10) + (1 - p_0)\, u(\$0) \ =\ p_0.$$

If the agent reports $p_0 = 0.6$, then $u(\$5) = 0.6$.

4. Cardinal vs. Ordinal Utility

Unlike ordinary utility, vNM utility carries **cardinal** information: differences between utilities are meaningful (they encode trade-offs across probabilities).

Example

With $A = \{a, b, c\}$ and $a \succ b \succ c$, there exists $\alpha \in [0, 1]$ with $b \sim (\alpha \circ a,\ (1 - \alpha) \circ c)$. Hence

$$u(b) = \alpha\, u(a) + (1 - \alpha)\, u(c) \quad \Longrightarrow \quad \frac{u(a) - u(b)}{u(b) - u(c)} = \frac{1 - \alpha}{\alpha}.$$

The *ratio of utility differences* reflects the preference intensity — a purely cardinal statement.

A monotone transformation that does *not* preserve such differences (like squaring) is no longer a vNM representation.

Example

Take the table $u(\$0) = 0$, $u(\$5) = 0.6$, $u(\$10) = 1$. Compare to $\tilde{u} = u^2$:

	$\$0$	$\$5$	$\$10$	$u(\$5) - u(\$0)$	$u(\$10) - u(\$5)$
$u(\cdot)$	0	0.6	1	0.6	0.4
$u(\cdot)^2$	0	0.36	1	0.36	0.64

With u, the difference between \$0 and \$5 is larger than between \$5 and \$10. With u^2 the order flips — so u^2 does **not** represent the same preferences.

Two vNM utility functions u and v represent the same preferences over $\mathcal{L}$ if and only if there exist $\alpha > 0$ and $\beta \in \mathbb{R}$ such that

$$v(g) \;=\; \alpha\, u(g) + \beta.$$

Proof. Positive affine transformations preserve utility differences:

$$\frac{(\alpha\, u(a) + \beta) - (\alpha\, u(b) + \beta)}{(\alpha\, u(b) + \beta) - (\alpha\, u(c) + \beta)} = \frac{u(a) - u(b)}{u(b) - u(c)}.$$

Hence the indifference $b \sim (\alpha \circ a, (1-\alpha) \circ c)$ is preserved, and so is the expected utility property. The converse is obtained by setting $\alpha = (v(a_1) - v(a_n))/(u(a_1) - u(a_n))$ and matching the endpoints. ■

vNM utility is **neither** fully cardinal (we cannot speak of absolute utility levels) **nor** fully ordinal (monotone transformations break the expected utility property). It is exactly as informative as the ratios of utility differences.

5. Risk Aversion

Restrict attention to lotteries with monetary outcomes: $A = \mathbb{R}_+$, $g = (p_1 \circ w_1, \ldots, p_n \circ w_n)$. Assume $u'(w) > 0$ (more money is better).

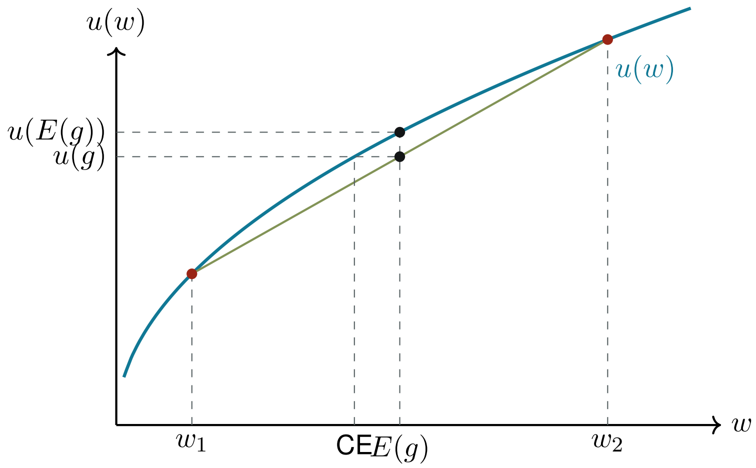
$$E(g) \;\equiv\; \sum_{i=1}^{n} p_i\, w_i.$$

The key comparison is between the lottery g and the *certain* amount $E(g)$.

- **Risk averse:** $u(E(g)) > u(g)\;\; \forall\, g$ non-degenerate.
- **Risk neutral:** $u(E(g)) = u(g)\;\; \forall\, g$.
- **Risk loving:** $u(E(g)) < u(g)\;\; \forall\, g$ non-degenerate.

A vNM utility function $u : \mathbb{R}_+ \to \mathbb{R}$ exhibits risk aversion if and only if u is **strictly concave**.

Why? Jensen's inequality: $u\big(\sum p_i w_i\big) > \sum p_i\, u(w_i)$ exactly when u is strictly concave.



5.1. Certainty Equivalent and Risk Premium

The **certainty equivalent** of a lottery g is the sure amount of money CE such that

$$u(\text{CE}) = u(g).$$

The **risk premium** of a lottery g is

$$P = E(g) - \text{CE}.$$

For a risk-averse agent, $P > 0$ — she is willing to give up some expected money to avoid the gamble.

Example

Take $u(w) = \ln(w)$, which is strictly concave (so the agent is risk averse). Consider the lottery

$$g = \left(\tfrac{1}{2} \circ (w_0 + h),\ \tfrac{1}{2} \circ (w_0 - h)\right), \qquad 0 < h < w_0.$$

Expected value: $E(g) = \frac{1}{2}(w_0 + h) + \frac{1}{2}(w_0 - h) = w_0$.
Certainty equivalent: solve $\ln(\text{CE}) = \frac{1}{2}\ln(w_0 + h) + \frac{1}{2}\ln(w_0 - h)$:

$$\ln(\text{CE}) = \tfrac{1}{2}\ln\left((w_0 + h)(w_0 - h)\right) = \ln\sqrt{w_0^2 - h^2}.$$

Hence CE $= \sqrt{w_0^2 - h^2} < w_0 = E(g)$.
Risk premium: $P = w_0 - \sqrt{w_0^2 - h^2} > 0$.

6. Measuring Risk Aversion: Arrow–Pratt

How risk averse is an agent? Concavity of u is a yes/no property; we need a number.

$$R_a(w) \;\equiv\; -\frac{u''(w)}{u'(w)}.$$

$$R_r(w) \;\equiv\; -\,w\,\frac{u''(w)}{u'(w)} \;=\; w\,R_a(w).$$

The minus sign makes R_a positive for concave (risk-averse) utilities. Higher R_a implies a larger risk premium for any small fair gamble: more risk-averse individuals have lower certainty equivalents.

Example

For $u(w) = \ln(w)$:

$$u'(w) = \tfrac{1}{w}, \quad u''(w) = -\tfrac{1}{w^2} \implies R_a(w) = \tfrac{1}{w}, \quad R_r(w) = 1.$$

Constant relative risk aversion — the agent's behavior is scale-invariant in wealth.

Reference: Jehle & Reny, Advanced Microeconomic Theory*, Chapter 2.*

